

Introduction to CS & AI

Research Methodology & Ph.D. Survival Guide

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August 31, 2026



- 1 General Introduction
 - How to Become a Good AI Researcher

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Indeed “introduction to CS & AI” DOES NOT indicate that
we will go through all basic CS & AI materials step-by-step in this course!

Research Turns Questions into Trustworthy Evidence

Question → Evidence → Claim → Scrutiny → Reuse / Impact

A novel answer is not enough.

A good researcher builds a path that others can **inspect**, **challenge**, and **reuse**.

If you are still unfamiliar with

¹<https://cs61a.org/>

²<https://fa26.datastructur.es/>

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- UC Berkeley, CS 61B: Data Structures².
- (Even) UC Berkeley, CS 61C: Great Ideas in Computer Architecture (Machine Structures)³.

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Build Your Research Workflow

- **Foundations — MIT Missing Semester**

Shell, development environments, Git, debugging, packaging, and code quality

<https://missing.csail.mit.edu/>

- **Research engineering — CodeRefinery**

Collaboration, reproducible environments, FAIR data, testing, documentation, and automation

<https://hands-on.coderefinery.org/coderefinery/>

- **Empirical research design — Princeton COS 598D**

Experiments, causal claims, regression, benchmarks, and threats to validity

https://humans.cs.princeton.edu/teaching/spring2026_empirical_methods.html

A Researcher Needs More Than a Topic

Foundations	Mathematics, statistics, algorithms, and learning theory
Build	Software, systems, data, security, and reliable infrastructure
Discover	Problem formulation, literature, experiment, and evaluation
Across all	Communication, collaboration, ethics, and societal impact

Competence = knowledge + practice + professional responsibility

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 - responsible use of AI

How to Become a Good AI Researcher

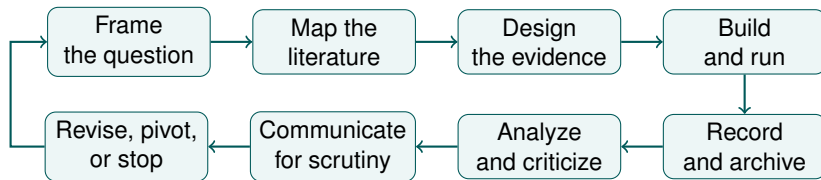
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- Research competencies (what we will introduce in this course: **A Survival Guide to a Ph.D.**)
 - effective communication strategy
 - research reasoning and experimental design
 - reproducible research engineering
 - responsible use of AI
- Practice these skills and make steady progress in your research

Research Is a Loop, Not a Checklist



AI can accelerate every step; **you still own every decision.**

A Result Is Finished Only When It Can Be Audited

Evidence

- Claims match experiments or proofs
- Baselines are fair and justified
- Multiple runs quantify uncertainty
- Failure cases and limitations are visible

Trace

- Data provenance, license, and splits
- Environment, dependencies, and compute
- Exact command or workflow to reproduce
- Code, data, and artifacts where possible

Use AI as an Assistant, Not as an Alibi

Verify Check facts, citations, summaries, code, and analysis against primary evidence.

Disclose Record substantial use: tool, version/date, purpose, and its effect on the work.

Protect Do not upload unpublished, personal, confidential, or review material without safeguards.

Own Humans remain responsible for scientific judgment, authorship, and every final claim.

Never outsource peer review or research accountability.

Study Agents as Systems, Not Chatbots

System Model + tools + memory + environment define what the agent can perceive and do.

Method Plan → act → observe → revise turns a model response into an interactive process.

Evidence Measure task success, trajectory quality, cost/latency, robustness, and safety—not just the final answer.

A credible agent claim connects architecture, behavior, and evaluation.

Course Schedule

Week	Date	Topics
1	2026. Sep. 01	How to communicate
2	2026. Sep. 08	How to do presentation
3	2026. Sep. 15	How to be a good AI researcher (I): doing research I
4	2026. Sep. 22	How to be a good AI researcher (II): productivity and career
5	2026. Sep. 29	How to be a good AI researcher (III): academic paper writing and peer reviews
6	2026. Oct. 13	Sharing the experience of writing excellent academic papers and rebuttal
7	2026. Oct. 20	Practice course I
8	2026. Oct. 27	Practice course II

How to communicate

- 1 A General Guide
 - Why Communication Matters?
 - The 7 **C**'s of Communication
- 2 How to Communicate With Your Collaborator?
 - How to Work With Your Advisor Effectively
 - How to Share Progress With Your Mentors/Collaborators?
 - How to Work With a Busy Advisor?
 - How to Work With Your Senior Advisor(s)?
- 3 How to Ask Questions The Smart Way (From CS Perspective)?
 - Before You Ask
 - When You Ask
- 4 How to Do Presentation

How to do presentation

- 1 Reminder: Principle of Effective Communication
- 2 How to Present—A General Guideline
 - A General Guide
 - Before the Talk / Preparing Your Talk
 - The Beginning of the Talk
 - The Body of the Talk
 - The End of the Talk
- 3 Others
 - How to Handle Questions in a Presentation?
 - How to Present a Line Plot?
 - How to Make a Research Poster?
 - How to Present a Poster at a Conference?
 - How to Present a Paper in Theoretical Computer Science: A Speaker's Guide for Students?

How to be a good AI researcher (I): doing research

- 1 Course Logistics
- 2 Recitation
- 3 How to Do Research
 - The Illustrated Guide to a Ph.D.
 - 10 Easy Ways to Fail a Ph.D.
 - How to Make Steady Progress?
 - How to Keep Track With the Literature?
 - How to Read Papers?
 - How to Come up With Research Ideas?
 - How to Do Experiments?
 - How to Create More Impact
- 4 Concluding Remarks

How to be a good AI researcher (II): productivity and career

1 Recitation

2 How to Do Research

- More on How to Read Papers
- 12 Resolutions for Grad Students
- How to Manage Your Time?
- How to Be Productive?
- Tips for Work-Life Balance (WLB)
- Others Career Tips

How to be a good AI researcher (III): writing and peer review

- 1 How to Write a Great Research Paper?
 - A General Guideline
 - How to Write the Introduction?
 - How to Write Papers That Are Easy to Read?
 - Tips to Create a Good Table
- 2 How to Write a Rebuttal for a Conference?
- 3 Summary

From Advice to Practice

Oct. 13 Writing excellent academic papers and a strong rebuttal

Oct. 20 Practice course I

Oct. 27 Practice course II

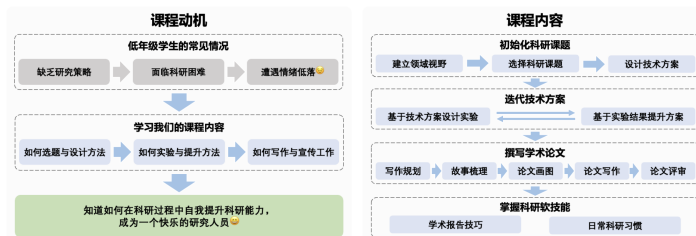
The course ends with practice:
select evidence, build a story, present it, and respond to questions.

Parallel course

GAMES003: 科研基本素养

2024 年秋季学期（在线直播）

课程介绍



本课程为初学者展示了一条全面的学术研究路径，旨在引领大家以系统性的方法探索计算机视觉和图形学领域的科学前沿。我们将指导大家从建立领域视野，到选择科研课题、设计技术方案，再到设计实验、优化方案、管理论文投稿、设计论文图表、撰写论文、自我评审与rebuttal，以及学习做学术报告的技巧，覆盖了科研过程中的每一个关键步骤。课程中，我们将结合具体案例，分享科研经验，同时鼓励学生提出问题，以实现具象的科研素养教学。

具体课程内容请参见课程大纲。

<https://pengsida.net/games003/>

Thanks & Question Time!